

Younghoon Kang

Undergraduate Researcher — On-Device AI

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EDUCATION

Seoul National University of Science and Technology (SeoulTech) 3rd-year undergraduate
B.S. in Computer Science and Engineering
GPA: 3.59 / 4.5

PROJECTS

coad-mini — Continual Learning for Streaming Video Action Recognition 2026
[code](#) · [live demo](#) · [project page](#) · [HF Space](#)

- Built an end-to-end continual-learning pipeline for streaming video action recognition (**CLIP ViT-B/32 → GRU → A-GEM**) on a 48-class Something-Something V2 subset, learning new actions over 8 sequential stages without catastrophic forgetting.
- A-GEM delivered **+14.1 %p average accuracy** and **+15.7 %p first-stage forgetting protection** over the sequential baseline (5-seed validated); multi-seed ablations showed **data scale, not architecture, is the dominant lever**.
- Added **few-shot enrollment** of new classes (live model hot-swap) and **OOD detection**, and shipped a **FastAPI** real-time webcam/video demo (Docker, Hugging Face Spaces).

KBO Game Outcome Prediction — End-to-End Sabermetrics ML Pipeline 2026

- Built an end-to-end ML pipeline predicting KBO home-team win probability from **~60 engineered sabermetric features** (park-adjusted ERA, lineup OPS/RISP, bullpen penalty, rolling form) over **2,435 games (2023–2026)** collected via the Statiz API.
- Trained an **Optuna-tuned XGBoost / LightGBM / CatBoost ensemble** with Platt-scaling calibration and **time-series 5-fold CV**, reaching **ROC-AUC $\approx$ 0.70 / accuracy $\approx$ 0.66** on 2026-season validation.
- Diagnosed and fixed a feature-staleness leak via sample-weighted in-season rolling-stat blending, and automated the full daily fetch → retrain → predict → API-submit pipeline.

AI Car Racing — PPO from Scratch: Collapse Recovery, Obstacle Avoidance & Action-Space Ablation 2026
[code](#)

- Implemented **PPO from scratch** in PyTorch (actor-critic, CNN, GAE, rollout buffer) for Gymnasium **CarRacing-v3** from raw 96×96 pixels, and diagnosed/fixed a training **collapse** traced to 3 root causes (per-minibatch vs per-epoch KL early-stop, a broken cosine LR schedule, a silent dead reward term), recovering a stable agent at **clean mean ~667**.
- Built a custom **obstacle-avoidance environment** (pixel-visible obstacles, collision penalties, reproducible spawning) and fine-tuned the agent to avoid obstacles (seed-matched before/after: **-68 → +323**).
- Ran instrumented ablations (reward shaping, entropy, **2-action vs native 3-action**) showing the signed-throttle action wrapper is a useful inductive bias (clean **415 vs 229**).

TECHNICAL SKILLS

Languages & ML: Python, PyTorch, scikit-learn, XGBoost / LightGBM / CatBoost, Optuna, Hugging Face Transformers, CLIP / OpenCLIP
Data & Tools: pandas, NumPy, OpenCV, Gymnasium, TensorBoard, Git, Linux
Deployment: FastAPI, Docker, Hugging Face Spaces, GitHub Actions, pytest

AWARDS & SCHOLARSHIPS

KIAT–KSEA Korea–U.S. Advanced Technology Youth Exchange Scholar (3rd cohort) Aug–Dec 2025
National government-run exchange program: **Silicon Valley field trip** to leading tech companies/research institutes, STEM-leader webinars, mentoring, and an advanced-technology industry report.

Grand Prize, Inter-Club Hackathon — SeoulTech Nov 2024
As **president of the department software club**, planned and led a joint inter-club hackathon to the top prize.

Excellence Award, AI Ideathon — Jeju National University Dec 2025
Designed an AI-driven idea addressing regional challenges in Jeju; awarded the Excellence Prize.